Year 12 Maths - Vectors

Similar questions - Position vectors, ratios and perpendicular bisectors

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ relative to an origin O. The point P lies on AB with $AP : PB = 2 : 3$, and Q is the midpoint of OB.

(a) Show that $\overrightarrow{OP} = \frac{1}{5}(3\mathbf{a} + 2\mathbf{b})$. **[2]**

(b) Find $\overrightarrow{PQ}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. **[2]**

(c) The line QP is extended to meet OA extended at the point R. Show that $\overrightarrow{OR} = 3\mathbf{a}$ and hence find the ratio $OA : AR$. **[4]**

2 A and B have position vectors $\mathbf{a} = \binom{1}{2}$ and $\mathbf{b} = \binom{5}{4}$. The point P has position vector $\mathbf{p} = \binom{x}{y}$ and satisfies $|\mathbf{p} - \mathbf{a}| = |\mathbf{p} - \mathbf{b}|$.

(a) Describe, in terms of A and B, the set of all such points P. **[1]**

(b) Find a cartesian equation for this set of points. **[3]**

(c) Show that the midpoint of AB satisfies your equation. **[1]**

Question 1

(a) $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$ and $AP = \frac{2}{5}AB$, so $\overrightarrow{OP} = \mathbf{a} + \frac{2}{5}(\mathbf{b} - \mathbf{a}) = \frac{1}{5}(3\mathbf{a} + 2\mathbf{b})$

(b) $\overrightarrow{OQ} = \frac{1}{2}\mathbf{b}$. $\overrightarrow{PQ} = \frac{1}{2}\mathbf{b} - \frac{1}{5}(3\mathbf{a} + 2\mathbf{b}) = \frac{1}{10}(\mathbf{b} - 6\mathbf{a})$

(c) R lies on QP extended, so $\overrightarrow{OR} = \overrightarrow{OP} + k\overrightarrow{PQ}$ for some k, and also $\overrightarrow{OR} = \lambda\mathbf{a}$. $\frac{1}{5}(3\mathbf{a} + 2\mathbf{b}) + \frac{k}{10}(\mathbf{b} - 6\mathbf{a}) = \lambda\mathbf{a}$. Coefficient of $\mathbf{b}$: $\frac{2}{5} + \frac{k}{10} = 0$, so $k = -4$. Coefficient of $\mathbf{a}$: $\frac{3}{5} + \frac{24}{10} = 3$, so $\overrightarrow{OR} = 3\mathbf{a}$. Hence $OA : AR = 1 : 2$

Question 2

(a) The perpendicular bisector of AB (all points equidistant from A and B)

(b) $(x-1)^2 + (y-2)^2 = (x-5)^2 + (y-4)^2$, so $-2x - 4y + 5 = -10x - 8y + 41$, giving $8x + 4y = 36$, i.e. $2x + y = 9$

(c) Midpoint $(3,3)$: $2(3) + 3 = 9$, which is true